MIL-T-24038D(EC)
28 October 1977
SUPERSEDING
MIL-T-24038C(EC)
1 November 1973
(See 6.7)

MILITARY SPECIFICATION

TRANSMITTER, RADIO AN/URT-24( )

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Radio Transmitter AN/URT-24( ) and associated accessories. This equipment is a digitally tuned transmitter intended for general purpose voice, continuous wave (CW) and radio teletypewriter communications, and may be used for surface ship, submarine and fixed shore installations. The equipment operates in the 2.0 to 30.0 megahertz (MHz) frequency range, and provides transmission of single sideband (selectable upper, lower and independent sideband), CW, AM (Compatible), radio teletypewriter and multichannel teletypewriter signals.

1.2 Operational requirements. Each unit shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in Naval ship and shore communications service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.i. (High-impact), Shipboard Machinery, Equipment And Systems, Requirements For (NAVY)
MIL-C-915	Cable And Cord Electrical, For Shipboard Use, General Specification For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-W-16878	Wire, Electrical, Insulated, High Temperature (NAVY)
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

MIL-F-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification For
MIL-C-22954	Control, Radio Set C-1138(·)/UR
MIL-A-23547	Antenna Coupler CU-937( )/UR
MIL-T-23645	Transmitter T-827( )/URT
MIL-M-38510	Microcircuits, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-167-1	Mechanical Vibration Of Shipboard Equipment (Type 1 - Environmental and Type 2 - Internally Excited)
MIL-STD-189	Rack, Electrical Equipment, 19-INCH and Associated Panels
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-883	Test Methods And Procedures For Microelectronics
MIL-STD-965	Parts Control Program

DRAWINGS

MILITARY

NAVAL SEA SYSTEMS COMMAND

RE-B2696865	Installation Control Drawings, Data list, For AN/URT-24( )

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. Equipments shall be in accordance with the provisions of MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3). The sample(s) shall be suitable for complete evaluation of mechanical and electrical form, design and performance. It shall be of final mechanical and electrical form, employ standard or nonstandard (when approved by the procuring activity) parts, and be completely representative of final equipment.

3.1.2 Parts, materials and processes. Parts, materials and processes shall be in accordance with MIL-E-16400, except as otherwise specified herein. Particular attention shall be given to the acceptable and unacceptable materials of MIL-E-16400.

3.1.2.1 Selection of parts, materials and processes. Selection of parts, materials and processes shall be in accordance with MIL-E-16400. Standard parts are as indicated in MIL-E-16400.

3.1.2.2 Approval of nonstandard parts. Requests for approval for the use of nonstandard parts shall be submitted in strict accordance with the requirements of MIL-STD-965.

3.1.2.2.1 Prescreening of approval requests. Prescreening of nonstandard parts approval requests shall be accomplished in accordance with MIL-STD-965.

3.1.2.2.2 All nonstandard parts shall have been documented in accordance with the requirements of MIL-STD-965, and approved, prior to conducting the first article tests required by 4.3.

3.1.2.2.3 All approved nonstandard parts shall be subjected to quality conformance inspection, in accordance with the pertinent component specification, as specified in MIL-E-16400. Inspection shall be as specified by the procuring activity upon submission, otherwise Group A only in accordance with the pertinent component specification.

3.1.2.2.4 If any Government furnished equipment (GFE) is provided, it may include nonstandard parts. The equipments furnished in accordance with this specification shall employ standard parts or approved nonstandard parts. Nonstandard part approvals previously granted under earlier contracts for equipments represented by the GFE shall not be construed as being extended to equipments furnished in accordance with this specification. Nonstandard parts cited in Government furnished drawings shall not be construed as being approved for use in equipments furnished in accordance with this specification.

3.1.2.3 Individual selection. The performance of the equipment shall not be dependent on the selection of individual parts, unless such selection is specifically approved by the procuring activity. If such individual part selection is approved, sufficient information and data shall be included as is necessary in all applicable technical publications and engineering drawings to enable designated repair activities to perform such selection.

3.1.2.4 Electron tubes. All electron tubes shall be selected from among approved types in accordance with MIL-E-16400. Electron tubes shall be limited to applications corresponding to circuits of the GFE.

3.1.2.5 Microelectronic devices. Selection and application of microelectronic devices shall comply with Requirement 64 of MIL-STD-454 requiring MIL-M-38510 Class B devices. When deviations to these requirements are necessitated due to availability or cost considerations, hermetically sealed devices inspected and tested to the requirements of MIL-STD-883 Class B may be used.

3.1.2.6 Semiconductor devices. All semiconductor devices shall be made of silicon. This was not a requirement for early models of this equipment, and may require redesign.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

3.1.3 Production equipment. Production equipments shall be identical to the equipment that has been subjected to and passed the first article examinations and tests (see 3.1.1 and 4.3).

3.2 Government furnished material (GFM). Government furnished material, if any, shall be as specified (see 6.2).

3.3 Interchangeability. Interchangeability requirements, if any, shall be as specified in the contract or order (see 6.2).

3.4 Composition. The Radio Transmitter AN/URT-24( ) shall consist of one each of the following:

a. Transmitter (Exciter) T-827( )/URT in accordance with MIL-T-23645
b. Radio Frequency (RF) Amplifier AM-3007( )/URT (with 115 VAC power supply provided and installed)
c. Shock and Vibration Mounting Assembly MT-3761( )/UR (see 3.8.1)
d. Antenna Coupler CU-937( )/UR, in accordance with MIL-A-23547
e. Interconnection Box
f. Set of cables (assembled) for interconnection of Transmitter, RF Amplifier and Interconnection Box units
g. Set of mating connectors for all receptacles for which interconnection cables of f are not furnished
h. Set of mounting brackets for stack mounting the equipment units

3.4.1 Accessories. In addition to the basic transmitter of 3.4, the following accessory items shall be furnished in the quantities specified (see 6.2):

a. Equipment plug-in assembly extension cables, as follows:
   1. Cables specified in MIL-T-23645
   2. RF Amplifier Power Supply Subassembly cable (one each)
   3. RF Amplifier DC to DC Converter Subassembly cables (two each)
   4. RF Amplifier APC/PPC Subassembly Cables (two each)

b. Set of printed circuit board (PCB) extender cards, one for each PCB in the equipment

3.5 Frequency range. The frequency (tuning) range of the equipment shall be from 2.0 to 30.0 MHz. Each equipment shall be capable of adjustment to any frequency from 2.0 from 29.9999 MHz in 100 Hertz (Hz) increments.

3.5.1 Method of tuning. Frequency selection over the complete frequency range of the equipment shall be accomplished in increments of 100 Hz, where changing the setting of the digital tuning knobs and loading the antenna coupler is all that is required to tune the equipment (see 3.15.1.6). Frequency tuning of the RF Amplifier shall be accomplished automatically when the digital tuning knobs of the T-827( )/URT are switched. Loading of the antenna coupler shall be accomplished using the controls on the RF amplifier.

3.6 Primary power source. Each equipment shall be capable of operating and providing rated performance from a primary power source of 115 volts alternating current (VAC) ±10 percent, 48 to 420 Hz, single phase, with no degradation of performance by power line voltage variations (relative to the nominal) of ±10 percent or above specified power line frequency variations, with either of these variations acting separately or in combination. The primary of the AC power transformers shall be equipped with taps for 90, 95, 100, 105, and 110 percent of the nominal. The equipment shall be delivered with connection to the nominal 115 volt tap. The total power consumption at 115 volts, 60 or 400 Hz, shall not exceed 500 watts (W).

3.6.1 The equipment shall conform to the performance limits of operation under the conditions of transient voltage, transient frequency and power interruption as specified in MIL-E-16400 for shipboard equipment. The equipment shall include provisions to protect against transients generated when the tuning motors energize or deenergize.

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:48Z
Check the source to verify that this is the currentversion before use.

3.6.2 Electrical overload protection. Electrical overload protection shall be in accordance with MIL-E-16400, except as modified herein.

The following circuits shall be protected by a circuit breaker or fuses:

a. Each line of the alternating current (AC) power input
b. Circuits specified in MIL-T-23645

3.6.3 In a system installation, power shall enter the equipment's interconnection box and be distributed by way of the main interconnection cabling to the equipment's radio frequency (RF) Amplifier and T-827( )/URT Transmitter units (see 3.4).

3.6.4 In addition to the system interconnection power source, the T-827( )/URT Transmitter (see 3.4) shall be provided with a separate power input capability to permit independent unit operation when desired.

3.6.5 Filament voltage of the AM-3007 driver tube shall be held at 6.3 volts direct current (VDC) ±5 percent for all primary power source voltage variations described in 3.6, and for all environmental conditions in which operation is required.

3.7 Operational requirements.

3.7.1 Type of emission. Each equipment shall provide for the transmission of the following types of emission:

a. A1 Telegraphy without the use of modulating audio frequency (CW)
b. A3a Telephony, single sideband (SSB), reduced carrier
c. A3b Telephony, two independent sidebands (ISB)
d. A3h Telephony, single sideband (SSB), full carrier
e. A3j Telephony, single sideband (SSB), suppressed carrier
f. A7j Single or multichannel voice frequency telegraphy (RATT)
g. A9b Two independent sidebands, combination of telephony and telegraphy (ISB/RATT)

3.7.2 Modes of operation. The equipment shall be capable of transmitting the emissions specified in 3.7.1 by selection of one of the following modes of operation:

a. LSB - Lower sideband
b. USB - Upper sideband
c. ISB - Independent sideband
d. AM - Amplitude modulation (compatible)
e. CW - Continuous wave telegraphy
f. RATT - Single or multichannel telegraphy
g. ISB/RATT - Lower sideband audio/single channel telegraphy (USB)

3.8 Equipment arrangement. Normal installation arrangement for the AN/URT-24( ) shall be as follows, starting from the bottom mounting surface:

a. Shock and Vibration Mounting Assembly MT-3761( )/UR
b. Transmitter T-827( )/URT
c. RF Amplifier AM-3007( )/URT

3.8.1 Shock and vibration mounting assembly MT-3761( )/UR. Each AN/URT-24( ) shall be delivered with the necessary brackets and hardware required to attach the units to the mounting assembly.

3.8.2 The Antenna Coupler CU-937( )/UR shall be installed as indicated in MIL-A-23547.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the currentversion before use.

3.9 Radio set system. The contractor shall be responsible for demonstrating complete system integration and system operation for the equipment to be delivered in accordance with the requirements of this specification. Equipment provisions for external cabling to the standard Navy shipboard radio remote control systems shall be required. Standard Navy cables and related information are contained in MIL-C-915. Mating connectors for all external cabling which require manufacture by the installing activity shall be furnished with each equipment. Interconnection cabling, plugs and receptacles for the interconnection of the T-827( )/URT, AM-3007( )/URT and interconnection box units shall be supplied with each AN/URT-24( ) equipment. Additional information pertaining to installation of the equipment may be found in the Installation Control Drawing, RE-B2696865.

3.9.1 Radio remote control system. The equipment shall contain compatible provisions for remote control of the radio transmitter in the standard Navy radio remote control system (not frequency control) using a maximum of three radio set Control Units C-1138( )/UR, in accordance with MIL-C-22954. Continuous simultaneous key-down operation from only two C-1138( )/UR units shall be required. Start-stop provisions are not required with use of C-1138( )/UR control units.

3.10 Dimensions and weights. The maximum dimensions and weights of the equipment shall not exceed those shown in TABLE I.

TABLE I. Dimensions and weights.

Units	Width	Depth 1/	Height	Weight
	Inches (cm)	Inches (cm)	Inches (cm)	Pounds (kg)
Transmitter	17-3/8 (44.13)	18-1/2 (47.0)	7 (17.75)	75 (34.0)
RF Amplifier (with 115 VAC power supply installed)	17-3/8 (44.13)	16-1/2 (42.5)	7 (17.75)	80 (36.4)
Shock/Vibration Mount Assembly	19-3/4 (50.0)	16-3/4 (43.2)	3-5/8 2/ (9.35)	17 (7.52)
Antenna Coupler	12 (30.5)	22 (length) (56.0)	12 (30.5)	30 (13.6)

1/ Exclusive of rear connectors

2/ Maximum unloaded height of bedplate

3.11 Environmental service conditions. The equipment shall be capable of continuous, reliable operation, within the performance limits specified herein, under those environmental conditions of MIL-E-16400 listed herein. The equipment shall operate within tolerances throughout the corresponding test specified in section 4 without alignment or adjustment, other than external controls normally employed for operation of the equipment.

a. Operating temperature range (Range 4)
b. Non-operating temperature range
c. Humidity (0 to 100 percent)
d. Salt fog (spray) (sheltered)
e. Shock
f. Vibration
g. Inclination (60 degrees)

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

3.12 Orientation. The equipment shall be capable of storage in any orientation.

3.13 Reliability.

3.13.1 Quantitative reliability requirement. The specified mean-time-between-failures (MTBF) (θ_0 as defined by MIL-STD-781) shall be 500 hours for this equipment.

3.13.2 Environmental burn-in. All equipments shall be subjected to an environmental burn-in for a cumulative operating time of at least 24 hours. The equipment shall be operated at a nominal 100 watts output at any frequency within the specified frequency range. No adjustments of the equipment, other than operating adjustments, shall be allowed.

3.14 Basic conditions. The following basic conditions shall apply as pertinent to the performance requirements specified herein except as modified by a particular test or test procedure contained in this specification. These conditions are specifically applicable to the measurement of the output power, spurious responses, intermodulation distortion, bandwidth and carrier suppression in the transmitter unit; and the measurement of the power output, harmonic output, and distortion in the RF amplifier unit.

3.14.1 Standard test conditions.

3.14.1.1 Primary power. The primary power test condition shall be 115 VAC, 60 Hz, single phase.

3.14.1.2 Ambient temperature. The ambient temperature shall be between 20° and 30°C.

3.14.1.3 Relative humidity. The relative humidity shall be less than 95 percent.

3.14.1.4 External/internal frequency standard switch. The FREQ STD switch shall be adjusted to its INT/COMP position.

3.14.1.5 Internal frequency standard. The internal frequency standard adjustment control shall be adjusted as required to ensure correct setting of the output frequency.

3.14.1.6 Transmitter RF output. Transmitter RF output shall be measured with a 50 ohm resistive load connected to the RF output terminal through a 1.5 meter (m) (five foot) length of RG-58 cable.

3.14.1.7 RF amplifier RF input. All RF amplifier RF inputs shall be applied to the unit from a standard source impedance (see 6.6.4).

3.14.1.8 RF amplifier bias. The RF amplifier bias shall be adjusted to ensure that the correct operating point has been set for the tubes.

3.15 Mechanical design and construction. Mechanical design and construction furnished in accordance with this specification shall conform to all applicable requirements of MIL-E-16400, to the extent of being compatible with the GFE.

3.15.1 General requirements.

3.15.1.1 Construction. Construction shall conform to the requirements of MIL-E-16400. The transmitter and RF amplifier shall consist basically of a frame, a front panel, a case and a group of replaceable plug-in assemblies. Retractable frame slides shall be provided on the transmitter and RF amplifier that will permit the frame, when withdrawn from the case, to be oriented ±90 degrees from its normal horizontal position . Locks shall be provided to maintain the frame in the desired position. The frame shall include the power supply (transmitter), receptacles for insertion of the plug-in assemblies, and required control functions. The front panel shall be securely mounted to the frame, with two handles provided, one mounted on each side of the front panel and each capable of supporting the entire weight of the unit. The case shall be of a molded construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

3.15.1.2 Case. The case shall be made of aluminum. The case shall be of an extruded construction, the same as that of the GFE. A retractable cable and a frame mating receptacle shall be provided to enable operation of the equipment when the frame is withdrawn from the case and oriented ±90 degrees from its normal horizontal position. Additional accessory cables shall not be required for operation when the equipment is withdrawn from the case. The retractable cable shall be installed so that it does not jam when the frame is inserted in or withdrawn from the case.

3.15.1.2.1 Rack mounting. The equipment shall be designed to permit standard rack mounting, in accordance with MIL-STD-189, when required. The equipment shall be designed and constructed so that when mounted in a standard rack, and not subject to shock, no other support or reinforcing members will be required to preclude deformation of any part of the equipment. The equipment's technical manual shall also include a sample fabrication drawing for the manufacture, by an installing activity, of suitable mounting brackets that will permit mounting of the units in a standard mounting rack, providing the notching conforms to MIL-STD-189.

3.15.1.3 Enclosure. The enclosure for the equipment shall be in accordance with MIL-E-16400. The degree of enclosure of the equipment shall be dripproof (45 degrees), except that the CU-937( )/UR shall conform to the requirements of MIL-A-23547.

3.15.1.3.1 External connections. External connections shall be the same as in the GFE.

3.15.1.4 Bonding and grounding. The equipment shall meet the requirements of MIL-E-16400, except as otherwise specified herein.

3.15.1.4.1. Case bonding. Case bonding and grounding shall be accomplished by preparing mounting surfaces to provide metal contact between the case and mounting bracket, and between the mounting bracket and shock mount. The area around each threaded insert of the case used to secure the mounting bracket to the case shall be free of paint and chemically treated to preserve the metal and provide a ground. Mounting brackets and the shock mount used with the equipment shall be similarly treated to insure bonding and grounding.

3.15.1.4.2 Grounding stud. The stud shall be a pressed-in captive insert made integral with the case and constructed of austentic corrosion-resistant steel. The captive insert shall be flush with the outside surface of the case and shall contain a blind threaded hole to prevent the entrance of moisture into the case. A 25.4 millimeter (mm) (one-inch) diameter area around the threaded insert shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.15.1.5 Corrosion protection, corrosion resisting treatments and painting. Corrosion protection and corrosion-resisting treatments shall be in accordance with MIL-E-16400.

3.15.1.5.1 Painting. Painting shall be in accordance with MIL-E-16400, except that the interior surfaces of each metal equipment enclosure shall be painted a dull black to improve thermal efficiency.

3.15.1.6 Frequency selection system. The frequency selection system shall be in accordance with MIL-T-23645.

3.15.1.7 Fastener hardware. Fastener hardware shall conform to the requirements of MIL-E-16400, except as otherwise specified herein. All screws, nuts, bolts, and other fastener hardware shall be of austenitic corrosion-resistant steel, except in those applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on equipment front panels and electronic plug-in assemblies. Cadmium plating is not acceptable on the captive-type bolts used on the equipment front panels and electronic plug-in assemblies.

3.15.1.8 Controls, indicators and panel layouts. Controls, indicators and panel layouts, including lights, receptacles, switches, meters (including functions) and fuseholders, shall be the same as furnished in the GFE, unless otherwise specified herein. A dimmer control for the equipment panel lights is not required.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the currentversion before use.

3.15.1.9 Electrical and radio frequency (RF) connectors. Electrical and radio frequency connectors shall be provided in accordance with and shall conform to the requirements of MIL-E-16400. All non-RF connectors shall be of the solder-lug type; crimped pin types shall not be used. Mating connectors for non-RF connectors shall be provided with cable clamps.

3.15.1.9.1 RF input and output connectors. The coaxial radio frequency connectors for the external frequency standard input and the antenna output shall be type N. The coaxial radio frequency connector for the internal frequency standard output shall be type BNC. The receptacles shall be mounted at the rear of the equipment cases and shall be suitably labeled. The center pin of the Type N connectors shall be made captive in all directions.

3.15.1.9.2 Assembly (module) connectors. RF inserts, if used, shall be constrained from nonperpendicular movement to ensure proper mating.

3.15.1.10 Time meter. A time meter is not required.

3.15.1.11 Locking devices. Locking devices on external operating controls are not required unless otherwise specified herein.

3.15.1.12 Panel lights. Panel lights shall be easily replaceable from the front panel without requiring removal of the frame from its case.

3.15.1.13 Motors. Motors used in the equipment shall comply with the requirements of MIL-E-16400, to the extent of compatibility with the Government furnished equipment.

3.15.1.14 Fuses. Fuses shall be in accordance with MIL-E-16400, except that extra fuses are not required. Fuses shall be mounted on the front panel in blown fuse indicating fuseholders which conform to MIL-F-19207. Spare fuseholders are not required.

3.15.1.15 Filters. Filters used in the sideband selection process shall be electromechanical or crystal filter type.

3.15.1.16 Head telephones and plugs. Head telephones and telephone plugs are not required to be furnished by the contractor.

3.15.1.17 Hand keys and plugs. Hand keys (CW) and plugs are not required to be furnished by the contractor.

3.15.1.18 Assembly identification. All assemblies to which the interchangeability requirements apply (see 3.3) shall be provided with the appropriate assigned National Stock Number, Manufacturer's Part Number (including revision indicator), Federal Supply Code Number, and Serial Number. This information shall be inscribed on the baseplate of the assembly.

3.15.1.18.1 Dust covers. The dust covers of all assemblies employing printed circuit boards shall indicate the location of all test points and all adjustments accessible through the dust cover.

3.15.1.19 Airborne and structureborne noise. The airborne and structureborne noise requirements of MIL-E-16400 are not applicable.

3.15.1.20 Accessibility. In addition to the requirements of MIL-E-16400, the circuit mounting boards, components, and tubes shall be easily accessible for maintenance, repair or replacement. This requirement shall be satisfied to the extent of being compatible with the GFE. Components which require adjustment prior to start of operation shall be accessible without having to remove dust covers or other coverings.

3.15.1.21 Test provisions. Test points shall be so provided and located as to permit the isolation of malfunctioning parts or subassemblies within any major unit or assembly, without the unsoldering of any interconnection or wiring lead. Test points provided shall be identical with those extant in the GFE.

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:48Z
Check the source to verify that this is the current version before use.

3.15.1.21.1 AM-3007( )/URT test points. Test points shall also be provided in the AM-3007( )/URT to measure the grid inputs to the tubes, and at the input to the RF bridge circuit.

3.15.1.22 Circuit alignment. Resonant circuits shall be provided with adjustable trimmers, or equally effective means, to permit accurate alignment. The trimmers shall provide sufficient means of adjustment to compensate for the effects of normal aging of electron tubes or other parts. If shrinkable tubing is used on variable components, it shall permit necessary adjustments to be made without having to replace the tubing.

3.15.1.23 Factory adjustment controls. The design of the equipment shall not include factory or sealed adjustment controls, unless specifically approved by the procuring activity. Where these controls are approved, a complete description and instructions for adjustment of such controls shall be included in the associated technical publications.

3.15.1.24 Shock, vibration, and inclination.

3.15.1.24.1 Shock. The equipment shall be capable of withstanding the shock test for Grade A, deck mounted, Class II, lightweight, Type A equipment as specified in MIL-S-901.

3.15.1.24.2 Vibration. The equipment shall be capable of withstanding the Type I vibration test of MIL-STD-167-1. During vibration testing, the equipment shall conform to the requirements of 3.16.1.4.2. The nameplate requirements of MIL-STD-167-1 are not applicable.

3.15.1.24.3 Inclination. The equipment shall meet the inclination requirements of 3.11 and MIL-E-16400.

3.15.1.24.4 Shock and vibration mount distortion. The shock and vibration mounting assembly shall show no deformation, bottoming or leaning on completion of the shock, vibration or inclination tests.

3.15.1.24.5 Shock, vibration and inclination tests shall be performed using the shock a vibration mounting assembly as specified in 3.8.a.

3.15.1.25 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the requirements of MIL-E-16400, to the extent of being compatible with the GFE.

3.15.1.26 Soldering. Soldering shall conform to the requirements of MIL-E-16400.

3.15.1.27 Welding. Welding shall conform to the requirements of MIL-E-16400.

3.15.1.28 Attachment of wires and leads. Attachment of wires and leads shall conform to the requirements of MIL-E-16400.

3.15.1.29 Internal wiring and cabling. Internal wiring and cabling shall conform to the requirements of MIL-E-16400. All wiring shall be TFE insulated. Hook-up wire shall conform to MIL-W-16878, Type E.

3.15.1.30 Conductors. Conductors shall be in accordance with MIL-E-16400, except that the number and purpose of conductors shall be the same as those used in the GFE.

3.15.1.30.1 Conductor identification. Identification of conductors shall meet the requirements of MIL-E-16400. The method used for identification shall be consistent throughout the equipment.

3.15.1.31 Identification and marking. Identification and marking of all units, assemblies, electrical and moving mechanical parts, and any mechanical parts required to be removed for normal maintenance, shall be in accordance with MIL-E-16400. Hertz or Hz shall be used for all indications of frequency on hardware and associated software. Reference designators shall be the same as used in the GFE to the maximum extent practicable.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

3.15.1.32 Part replacement. Part replacement shall conform to the requirements of MIL-E-16400.

3.15.1.33 Quartz crystal units. Quartz crystal units shall conform to the requirements of MIL-E-16400.

3.15.1.34 Gears and cams. Gears and cams shall conform to the requirements of MIL-E-16400, except that only metal gears shall be used.

3.15.1.35 Extender cables. Extender cables for each assembly listed in 3.4.1a shall be furnished as required (see 6.2). Each completed cable shall be 609.6 mm (24 inches) long. These cables will be used to assist in maintaining the equipment.

3.15.1.36 Thermal design. Thermal design shall be in accordance with MIL-E-16400 to the extent of compatibility with the Government furnished equipment. A thermal survey shall be performed.

3.15.2 Transmitter (Exciter)T-827( )/URT. The T-827( )/URT shall meet the mechanical design requirements of MIL-T-23645.

3.15.3 Antenna coupler CU-937( )/UR. The CU-937( )/UR shall meet the construction requirements of MIL-A-23547.

3.15.4 Shock and vibration mounting assembly MT-3761( )/UR. The shock and vibration mounting assembly shall comply with the requirements of MIL-E-16400. The mount when attached to the AN/URT-24 shall permit the equipment to comply with the requirements of 3.15.1.24.

3.15.4.1 The assembly shall be streamlined to the extent that it shall have no sharp corners or edges. All corners shall be rounded.

3.15.5 Interconnection box. The interconnection box shall be capable of mounting to a flat vertical surface.

3.16 Detailed performance.

3.16.1 General (system) requirements. The following requirements are for operation using a dummy load in lieu of the antenna coupler.

3.16.1.1 Power output. The equipment shall be capable of transmitting 100 watts PEP (minimum) in the SSB mode (USB or LSB), 50 watts (minimum) in CW or RATT modes, and 25 watts carrier (minimum) in the AM mode, into a 50 ohm resistive load on a continuously keyed basis throughout the 2 to 30 MHz frequency range. The power output rating in the single sideband modes shall be based on the standard two-tone audio input.

3.16.1.1.1 Voltage standing wave ratio (VSWR) requirements. The equipment shall be capable of operation into a 50 ohm transmission line supplying an antenna system having a VSWR up to 3:1 without damage to the transmitter. The transmitter minimum forward power may be reduced with respect to the minimum levels specified in 3.16.1.1 by 2 decibels (dB) at a VSWR of 1.5:1, 4 dB at 2:1, and 6 dB at 3:1. The equipment shall not be damaged by operation into a transmission line with a VSWR greater than 3:1 including the conditions of open and short circuit.

3.16.1.1.2 Average power control (APC) power output leveling. The equipment shall be equipped with an APC system which operates to maintain the output power level within +1.5 dB of the minimum outputs of 3.16.1.1 over the frequency range from 2.0 to 30.0 MHz.

3.16.1.2 Frequency stability. The frequency stability of the transmitter system shall be as specified for the T-827( )/URT in MIL-T-23645.

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:48Z
Check the source to verify that this is the currenversion before use.

3.16.1.3 Modes of operation. Modes of transmitter operation shall be as specified in MIL-T-23645.

3.16.1.4 Intermodulation distortion. All intermodulation products of the equipment, when operated as a complete system, shall be reduced at least 30 dB below either tone of a two-tone envelope at a minimum of 100 watts PEP, from 2.0 to 30.0 MHz, produced by using the standard two-tone input (see 6.6.3).

3.16.1.4.1 APC modulation-following distortion. Intermodulation products shall be reduced at least 30 dB below either tone of a two-tone envelope at a minimum of 100 watts PEP with frequency separation of audio input tones reduced to 110 Hz.

3.16.1.4.2 Vibration test distortion. When tested in accordance with 3.15.1.24.2, any vibration related amplitude modulation shall be less than three percent of a single tone RF output of 25 watts. There shall be no breakup of the RF output signal.

3.16.1.5 Harmonic suppression. The suppression of harmonic outputs shall be referenced to the output produced by a single tone in the CW mode (50 watts nominal). Second harmonic output, when measured at 2.0000, 4.0000, and 8.0000 MHz shall be reduced at least 35 dB below the reference. Second harmonic output at all other test frequencies shall be at least 40 dB below the reference. All other harmonic outputs shall be at least 45 dB below the reference.

3.16.1.6 Carrier suppression. In the USB or LSB mode of operation with the internal carrier insertion switch in the full carrier suppression position, the carrier shall be suppressed a minimum of 50 dB below nominal 25 watts single tone RF output. In the ISB mode the carrier shall be suppressed a minimum of 45 dB below nominal 25 watts output in either sideband produced by a standard single tone audio input. With the carrier insertion switch in the 0 dB (full carrier), -10 dB, or -20 dB position, the carrier shall be within three dB of the selected level relative to a 25 watt single tone output in USB, LSB, or ISB modes.

3.16.1.7 Opposite sideband suppression. The level of the opposite sideband frequencies (measured at a frequency 1 kilohertz (kHz) or greater from the suppressed carrier) shall be suppressed at least 50 dB below a 1000 Hz single tone signal at 25 watts.

3.16.1.8 Spurious RF emissions. Spurious RF emissions, when measured across a standard 50 ohm resistive load at the transmitter RF output terminals, shall meet the following requirements.

3.16.1.8.1 Out-of-band emissions. All out-of-band spurious emissions, with the exception of harmonics of the primary RF band, specified in 3.16.1.5, shall be reduced at least 50 dB below a single tone output of 25 watts. Spurious emissions shall be reduced at least 50 dB below 25 watts for the condition of a keyed transmitter with no audio modulation (audio lines terminated in 600 ohms).

3.16.1.8.2 In-band products. Spurious emissions which fall within the intelligence band shall be reduced at least 40 dB below a single tone output of 25 watts. This requirement is for either a modulated, or keyed but unmodulated transmitter, and includes frequencies which are produced by modulation of the carrier by harmonics of the tone.

3.16.1.9 Noise. The transmitting system broadband noise output into a 50 ohm resistive termination shall be measured in the USB and LSB modes, and shall not exceed the following levels for the three kHz information band:

a. With the system in a key-down condition, audio inputs terminated in a 600 ohm resistance and with no signal input, the noise output shall be attenuated at least 45 dB below 25 watts average.
b. With the system in a key-up condition, audio inputs terminated in a 600 ohm resistance and with no signal input, the noise output shall not exceed -115 decibels referred to one milliwatt (dBm).

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

3.16.1.10 Power supply hum distortion. All hum distortion signals shall be suppressed a minimum of 45 dB with respect to an RF output of 25 watts produced by a standard single tone audio input at 1000 Hz.

3.16.1.11 Average power control (APC). An APC circuit shall be provided to maintain the average power output within safe limits when operated as a system, and to maintain the level of the reinserted carrier in an A3h (compatible AM) mode. Time constants of the system APC circuits shall be selected to minimize signal distortion (see 3.16.1.4.1).

3.16.1.12 Peak power control (PPC). A PPC circuit shall be provided to limit the power output peaks to a level sufficient to safeguard the equipment, including the Antenna Coupler CU-937( )/UR, from damage.

3.16.1.13 CW keying. Keying rates up to 32 baud shall be possible. The keyed wave shall be free from transients and splatter.

3.16.1.14 RATT keying. Keying speeds up to 80 baud, using total frequency shifts of either (a) 850 Hz, or (b) 170 Hz, centered on 2000 Hz, shall be possible. External RATT input shall be possible with a neutral input of 120 volts direct current (DC), 5 to 75 mA. RATT emission shall be in the upper sideband of the transmitter. This requirement has been modified relative to prior AN/URT-24( ) equipments and will require redesign.

3.16.1.15 Keying transient. There shall be no keying transient (signal burst) when the equipment is keyed and unkeyed.

3.16.1.16 Regeneration. There shall be no evidence of unintended oscillations or instability under any condition of equipment operation, including primary line voltage 10 percent higher than the nominal primary line voltage.

3.16.1.17 Oscillator radiation. Oscillator radiation shall conform to the Class IB requirements of MIL-STD-461.

3.16.1.18 Electromagnetic compatibility (EMC). The equipment shall meet the requirements of MIL-STD-461 for Class IB, C-E equipment. Tests CE01 and CE03 shall be performed at both 60 and 400 Hz power line frequencies.

3.16.1.19 Accelerated life. The equipment shall conform to the requirements of MIL-E-16400 for accelerated life.

3.16.2 Transmitter T-827( )/URT. The T-827( )/URT shall meet the performance requirements of MIL-T-23645.

3.16.3 RF amplifier AM-3007( )URT.

3.16.3.1 Power output. The minimum power delivered to a standard dummy load over the frequency range of 2.0 to 30 MHz shall be at least as specified in TABLE II, when driven by a source having the maximum power output capability indicated, for the mode of operation being measured.

TABLE II. Mode of operation.

	SSB(A3j)	Compatible AM(A3h)	CW/FSK (A1, A7j)	Independent sideband (A3b)
Input (max.)	250 mW PEP	62.5 mW carrier	125 mW carrier	62.5 mW PEP/ per sideband
Output (min.)	100 W PEP	25 W carrier	50 W carrier	25 W PEP/per sideband

Source: https://assist.dla.mil -- Downloaded: 202640-19T18:48Z
Check the source to verify that this is the currentversion before use.

3.16.3.1.1 The RF amplifier shall be capable of delivering these power outputs continuously into a 50 ohm load without damage to parts or tubes.

3.16.3.2 Frequency range and tuning. The frequency range shall be from 2.0 to 30.0 MHz. The RF amplifier shall accept coded channel information from the T-827( )/URT and automatically select the proper operating band and perform under the requirements of this specification.

3.16.3.3 Modes of operation. Modes of operation shall be as specified in 3.7.

3.16.3.4 Intermodulation distortion. All intermodulation products shall be at least 32 dB below either of two equal tones at RF frequencies corresponding to the standard audio two-tone input. This requirement shall be met at a minimum output of 100 watts PEP throughout the 2.0 to 30.0 MHz frequency range.

3.16.3.5 Harmonic output. The suppression of harmonic outputs shall be referenced to the output, produced by a single tone, of 50 watts average power. The second harmonic output shall be reduced at least 35 dB when measured at the 5 percent points of the 2, 4, and 8 MHz bands. Second harmonic output at all other test frequencies shall be at least 40 dB below the reference. All other harmonic outputs shall be at least 45 dB below the reference.

3.16.3.6 Broadband noise spectral density. Broadband noise output in a key-down condition measured in a 3 kHz bandwidth over the frequency range of 1 to 32 MHz shall not exceed -48 dBm.

3.16.3.7 Power supply hum distortion. All hum distortion products shall be suppressed a minimum pf 50 dB with respect to a single tone RF output of 25 watts.

3.16.4 Interconnection box. The interconnection box shall be the termination for all wiring used to interconnect the T-827( )/URT, AM-3007( )/URT, CU-937( )/UR, remote control system units and primary power sources. Within the box, all connnections will be made via terminal boards. All terminals and terminal boards will meet the safety requirements of MIL-E-16400.

3.16.4.1 Connectors. The interconnection box will be equipped with connectors that will accept the cables from the T-827( )/URT and AM-3007( )/URT. All connectors shall be located on the bottom of the interconnection box.

3.16.4.2 Cables. All cables between the interconnection box and associated units shall be at least 3.048 m (10 feet) in length.

3.16.4.3 Stuffing tubes. The interconnection box shall be equipped with stuffing tubes that will accept the cabling from the CU-937( )/UR, remote control system units and the primary power source. The stuffing tubes shall be located on the top of the interconnection box. The following stuffing tubes are required:

a. CU-937( )/UR
b. Primary power
c. Remote control system
d. Receiver audio input
e. Teletype input
f. Keyline closure

3.16.4.4 Additional functions. The interconnection box shall have provisions for the following functions:

a. Primary power line filtering
b. Audio transformer, 600 ohm center tapped, for coupling external audio to pins A. and B of the local handset connector of the T-827( )/URT
c. Protective fuse for remote 12 VDC
d. 820 ohm, 10 watt resistor for teletype loop termination
e. Circuitry for opening the keyline so that an external dry contact closure may enable/disable equipment operation. The purpose of this equipment is to ensure that the box may be safely opened when the equipment is connected to an antenna patch panel.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

3.16.4.5 Programming. The terminal boards associated with the CU-937( )/UR wiring shall be wired and jumpered to permit operation with a 10.7 m (35 foot) whip antenna.

3.16.5 Antenna coupler CU-937( )/UR. The CU-937( )/UR shall meet the performance requirements of MIL-A-23547.

3.17 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of product to be offered for acceptance; c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.2 General inspection. Inspection shall be as specified in MIL-E-16400 except as otherwise specified herein.

4.2.1 Classification of inspection. The methods of examination and testing of electronic equipment fall within the following classifications:

a. First Article inspection (see 4.3)
b. Quality conformance inspection
    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
    3. Environmental inspection (see 4.4.3)
c. Reliability testing (see 4.10)
d. Preparation for delivery inspection (see 4.11)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examinations and testing necessary to determine compliance with all applicable requirements of this specification, including the tests specified in TABLE III (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be made on every item offered for delivery. This inspection shall comprise such examination and tests which will prove the workmanship and reveal omissions and errors of the production process such as: functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls/adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examination and tests shown in Group A of TABLE III, subject to the conditions of 4.4.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

TABLE III. Examination and tests.

Examination or test	Requirement	Test	First article	QC INSP A	QC INSP B	QC INSP C
Surface Examination	3.1.2	MIL-E-16400	X	X		
Weight	3.10	MIL-E-16400	X	X		
Size	3.10	MIL-E-16400	X	X		
Parts & Materials	3.1.2	MIL-E-16400	X	X		
Finish	3.1.2	MIL-E-16400	X	X		
Marking	3.15.1.31	MIL-E-16400	X	X		
Workmanship	3.17	MIL-E-16400	X		X	
Weld	3.15.1.27	MIL-E-16400	X		X	
Mechanical Design	3.15	MIL-E-16400	X		X	
Bonding & Grounding	3.15.1.4	MIL-E-16400	X		X	
Corrosion Protection	3.15.1.5	MIL-E-16400	X		X	
Painting	3.15.1.5.1	MIL-E-16400	X		X	
Soldering	3.15.1.26	MIL-E-16400	X		X	
Conductors/Identification	3.15.1.30	MIL-E-16400	X		X	
System tests						
Power Output	3.16.1.1	4.9.1.1	X	X	X	
VSWR Protection	3.16.1.1.1	4.9.1.1.1	X		X	
Power Leveling	3.16.1.1.2	4.9.1.2	X	X	X	
Frequency Stability	3.16.1.2	4.9.1.3	X	X	X	
Intermodulation	3.16.1.4	4.9.1.4	X		X	
APC Distortion	3.16.1.4.1	4.9.1.4.1	X		X	
Harmonic Suppression	3.16.1.5	4.9.1.5	X	X	X	
Carrier Suppression	3.16.1.6	4.9.1.6	X	X	X	
Opp Sdbd Suppression	3.16.1.7	4.9.1.7	X	X	X	
Out-of-Band Spurious	3.16.1.8.1	4.9.1.8	X	X	X	
In-Band Spurious	3.16.1.8.2	4.9.1.8	X		X	
Noise	3.16.1.9	4.9.1.9	X		X	
Hum Distortion	3.16.1.10	4.9.1.10	X	X	X	
APC	3.16.1.11	4.9.1.11	X		X	
PPC	3.16.1.12	4.9.1.12	X		X	
CW Keying	3.16.1.13	4.9.1.13	X		X	
RATT	3.16.1.14	4.9.1.14	X		X	
Keying Transient	3.16.1.15	4.9.1.15	X		X	
Regeneration	3.16.1.16	4.9.1.16	X		X	
Oscillator Rad	3.16.1.17	4.9.1.17	X		X	
Remote Operation	3.9.1	4.9.1.18	X		X	
Steady State Voltage	3.6	4.9.1.23	X		X	
Steady State Frequency	3.6.1	4.9.1.23	X		X	
Transient Voltage	3.6.1	4.9.1.23	X		X	
Transient Frequency	3.6.1	4.9.1.23	X		X	
Power Interruption	3.6.1	4.9.1.24	X		X	
Power/Power Factor	3.6	4.9.1.23	X			X
Temperature Sequence	3.11	4.9.1.20	X			X
Low Temperature	3.11	4.9.1.20	X			X
High Temperature	3.11	4.9.1.20	X			X
Humidity	3.11	4.9.1.20	X			X
Salt Fog	3.11	MIL-E-16400	X			

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

TABLE III. Examination and tests. (Continued)

Examination or test	Requirement	Test	First article	QC INSP A	QC INSP B	QC INSP C
Accelerated Life	3.11	MIL-E-16400	X			X
Reliability	3.13	4.10				
Enclosure	3.15.1.3	4.9.1.22	X			X
Shock	3.15.1.24.1	4.9.1.21	X			X
Vibration	3.15.1.24.2	4.9.1.21	X			X
Inclination	3.15.1.24.3	4.9.1.21	X			X
Thermal Design	3.15.1.36	MIL-E-16400	X			
EMI	3.16.1.18	4.9.1.19	X			X
T-827 Tests	3.16.2	4.9.2	X		X	
AM-3007	3.16.3	4.9.3				
Power Output	3.16.3.1	4.9.3.1	X		X	
Intermodulation	3.16.3.4	4.9.3.2	X		X	
Harmonics	3.16.3.5	4.9.3.3	X		X	
Spectral Density	3.16.3.6	4.9.3.4	X		X	
Hum Distortion	3.16.3.7	4.9.3.5	X		X	
CU-937 Tests	3.16.3.5	4.9.4	X	X	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

4.4.1.1 Group A tests. The Group A tests shall be performed in accordance with the test methods of 4.9, except as modified by the following:

a. Power output - 2, 5, 10, 15, 20, 25 and 29 MHz bands
b. Intermodulation distortion - 2, 5, 10, 15, 20, 25 and 29 MHz bands and A3j mode
c. Harmonic suppression - 2, 15 and 29 MHz bands, second and third order
d. Carrier suppression and opposite sideband suppression - 2.1 MHz, USB and LSB
e. Spurious output - same as MIL-T-23645
f. Output signal-to-hum ratio - 2, 17 and 29 MHz bands
g. Frequency stability - 5 MHz
h. APC power output levelling - 2, 16 and 29 MHz bands

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified. This inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examination and tests shown in Group B of TABLE III. Production control inspection shall be conducted on inspection lots that have passed the production inspection of 4.4.1. Sampling shall be in accordance with MIL-STD-105, using inspection level S-3 with an acceptable quality level (AQL) of 6.5 percent for each attribute.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration and inclination. Environmental inspection shall include the examination and tests shown in Group C of TABLE III.

4.4.3.1 Sampling for inspection of equipment. For the first environmental inspection, one equipment from the first month's production, or from the first 50 units produced, whichever is less, shall be selected. For subsequent environmental inspections, one equipment from each successive 100, or one per month if less than 100 are produced monthly, shall be selected. EMI testing shall be conducted only on one of each successive 200 equipments. Tests shall be conducted on lots that have passed the production inspection of 4.4.1.

4.4.3.2 Noncompliance. If a sample unit fails environmental inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of products which were manufactured with the same conditions, materials, processes, and so forth. If the Government inspector considers that the corrective action will not enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.5).

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified, sample units which have been subjected to and passed production control or environmental inspections, or both, may be accepted on the contract provided that they are resubjected to and pass production inspection after repair of all defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

4.5 Test plan and test procedures. Test plans and test procedures shall be as specified (see 6.2.2).

4.6 Test reports. Test reports shall be as specified (see 6.2.2).

4.7 Test facilities, chambers and apparatus. Test facilities, chambers and apparatus shall be in accordance with MIL-E-16400.

4.8 Performance of test. Performance of tests shall be in accordance with MIL-E-16400 and as specified herein.

4.9 Test methods.

4.9.1 System tests.

4.9.1.1 Power output. Power output measurements shall be made for each mode of transmitter operation on each primary frequency band. Measurements shall be made at the 5, 50, and 95 percent points of each primary frequency band.

4.9.1.1.1 VSWR protection. Tests shall be conducted in the A1 mode with power amplifier RF output coaxial line VSWR of 1.5:1, 2:1, and 3:1 to show compliance with 3.16.1.1.1. VSWR protection shall be measured by terminating the RF amplifier output coaxial line with resistive loads to produce VSWR's of 1.5:1, 2:1, and 3:1. Measure forward and reflected power with a directional watt meter, and monitor the AM-3007 panel meter to detect combinations of VSWR, coaxial line length, and frequency that cause excessive power amplifier plate current.

4.9.1.2 Power leveling. Measurements shall be made to verify compliance with the requirements of 3.16.1.1.2.

4.9.1.3 Frequency stability. Frequency stability measurements may be made as part of the T-827( )/URT tests of 4.9.2.

4.9.1.4 Intermodulation distortion. Measurements shall be made in the A3j mode at the nominal end and midband frequencies for each primary frequency band. Intermodulation distortion shall be measured by applying a standard two-tone audio input (see 6.6.3) and monitoring the output developed across a standard dummy load (50 ohm) with a spectrum analyzer. APC shall be adjusted prior to beginning measurements, and no further adjustment shall be made.

4.9.1.4.1 APC modulation following distortion. Measurements shall be made at the T-827 high and low gain frequencies to show compliance with the requirements of 3.16.1.4.1.

4.9.1.5 Harmonic suppression. Harmonic outputs shall be measured at the end and mid-band frequencies of each primary frequency band.

4.9.1.6 Carrier suppression. Measurement shall be made at the nominal end and mid-band frequencies of the 2, 5, 10, 20 and 29 MHz bands.

4.9.1.7 Opposite sideband suppression. Measurements shall be made at the nominal end and mid-band frequencies of the 2,5, 10, 20, and 29 MHz bands.

4.9.1.8 Spurious RF emissions. Each primary frequency band shall be completely explored for evidence of spurious RF emissions which fall outside of the allowable limits of 3.16.1.8.

4.9.1.9 Noise. The transmitter noise output shall be checked on the 2, 16 and 29 MHz bands, on both USB and LSB.

4.9.1.10 Output signal-to-hum. Output signal-to-hum ratio shall be measured at 2.1, 17.1 and 29.9 MHz on both USB and LSB. Measurements shall be made for both 60 Hz and 400 Hz power line frequencies.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

4.9.1.11 APC output. The APC output shall be measured at the arithmetic mean frequency of each channel when an unmodulated RF carrier of sufficient amplitude to dissipate 25 watts average power in a 50 ohm standard dummy load has been applied to the RF amplifier input.

4.9.1.12 PPC output. PPC output shall be measured at the arithmetic mean frequency of each channel when an AM, 1000 Hz, 80 percent modulated RF signal, or a standard two-tone signal, is applied to the RF amplifier input.

4.9.1.13 CW keying. CW keying shall be tested by keying the transmitter to develop rated output and photographing the input and output of the RF amplifier as simultaneously displayed on an oscilloscope. All pertinent information regarding gain, frequency, time scale, and so forth shall be recorded on the photograph.

4.9.1.14 RATT. The equipment shall be tested to verify compliance with 3.16.1.14.

4.9.1.15 Keying transients. The equipment shall be tested to determine compliance with 3.16.1.15.

4.9.1.16 Regeneration. Each primary frequency band shall be carefully explored for evidence of regeneration, under the conditions of 3.16.1.16.

4.9.1.17 Oscillator radiation. Measurements shall be made to verify compliance with 3.16.1.17.

4.9.1.18 Local and remote operation. Tests shall be conducted using the C-1138( )/UR to verify compliance with 3.9.1.

4.9.1.19 Electromagnetic compatibility test. Compliance with 3.16.1.18 shall be verified by tests performed in accordance with MIL-STD-462.

4.9.1.20 Temperature and humidity. Tests shall be conducted in accordance with MIL-E-16400. The following measurements shall be made before and after the temperature and humidity tests, and during temperature operating tests at the end of the period of operation at 0°C and at the end of the period of operation at +50°C: power output in the A1 and A3j modes and intermodulation distortion at frequencies of 2, 5, 10, 15, 20, and 29.9 MHz; frequency stability, and synthesizer lock checks. During the temperature operating test the transmitter shall be keyed continuously to produce rated power output during operation at +50°C. During the last hour of the high temperature operating test, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the equipment under test shall be controlled to 50°C by radiation-baffled heating and cooling sources as required. This test is intended to simulate the worst case shipboard thermal environment (see 4.9.1.20.1). During humidity test, measurements shall be made at the end of the second and fifth cycles.

4.9.1.20.1 Chamber temperature control provisions. The sources for maintaining the chamber environment shall be arranged or suitably baffled so that no surface at a temperature other than the desired ambient is visible to the unit under test. The maximum allowable temperature difference between any two points in the chamber shall be 10°C. The chamber control provisions shall be capable of maintaining the temperature indicated by a selected chamber ambient sensor constant within ±1°C of any selected test temperature.

4.9.1.21 Shock, vibration, and inclination. Prior to and after completion of shock, vibration and inclination tests, the following tests shall be performed:

a. Power output measurements in accordance with 4.9.1.1 and the A1 and A3j modes for the 2, 17, and 29 MHz bands only
b. Synthesizer lock checks in accordance with MIL-T-23645
c. Intermodulation distortion in accordance with 4.9.1.4. During vibration tests, equipment operation and monitoring shall be in accordance with requirements of 3.16.1.4.2. Continuous key-down operation is required during all tests, unless otherwise specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

4.9.1.22 Enclosure. Tests shall be conducted in accordance with MIL-E-16400. Measurements of power output and frequency stability shall be made prior to and after testing. Measurements shall be within the limits established in 3.16.1.1 and MIL-T-23645.

4.9.1.23 Supply line voltage and frequency variation and transients. Tests shall be performed in accordance with MIL-E-16400. Reference power output and frequency measurements shall be made prior to start of testing. Variation of transmitter power output and frequency with line voltage and frequency shall be measured against the requirements of MIL-T-23645 for nominal line voltage of 115 volts AC, and for nominal line frequencies of 60 and 400 Hz. Following performance of the tests specified in MIL-E-16400, no permanent changes shall be produced in power output, frequency and gain.

4.9.1.24 Power interruption. In addition to the tests specified in MIL-E-16400, a test shall be conducted to verify operation with high voltage transients of three and one-half times nominal voltage but of short duration (less than one cycle). No permanent changes shall be produced in power output, frequency or gain as a result of this test.

4.9.1.25 Power and power factor. Tests shall be conducted in accordance with MIL-E-16400.

4.9.1.26 Accelerated life. Measurements of power output and frequency stability shall be made in accordance with 4.9.1.1 and 4.9.1.3 in the 2, 17, and 29 MHz bands to verify compliance with MIL-E-16400.

4.9.2 T-827( )/URT tests. The T-827( )/URT shall be subjected to the first article tests of MIL-T-23645. Only non-environmental tests are required.

4.9.3 AM-3007( )/URT tests.

4.9.3.1 Power output. Measurements shall be made for each mode at the nominal end frequencies and at the arithmetic mean of each frequency channel into which the overall frequency range of the RF amplifier has been divided.

4.9.3.2 Intermodulation distortion. Measurements shall be made at the nominal end frequencies and at the arithmetic mean of each frequency channel into which the overall frequency range of the RF amplifier has been divided.

4.9.3.3 Harmonic suppression. Harmonic outputs shall be measured at the end and mid-frequencies of the RF amplifier frequency bands.

4.9.3.4 Spectral density. Measurements shall be made to verify compliance with 3.16.3.6.

4.9.3.5 Hum distortion. Measurements shall be made to verify compliance with 3.16.3.7.

4.9.4 CU-937( )/UR tests. Tests of a complete AN/URT-24( ) including the CU-937( )/UR shall be conducted to verify compliance with MIL-A-23547.

4.10 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781. The reliability demonstration shall be performed after first article approval.

4.10.1 Reliability demonstration (qualification) test. Reliability demonstration testing shall be performed on at least six production units. Testing shall be performed at Test Level C and proceed to a decision in accordance with Test Plan IV of MIL-STD-781. The AN/URT-24 reliability demonstration shall be continued beyond the accept decision point as required to verify the separate T-827( )/URT and CU-937( )/UR reliability requirements of MIL-T-23645 and MIL-A-23547, respectively. All reliability demonstration requirements shall be successfully completed prior to delivery of any production equipments under the contract.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

4.10.1.1 Failure. Failure shall be any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements specified herein (for definition of failure, see 6.6.8). Any adjustments or alignments, after test initiation, shall be considered as a malfunction.

4.10.1.2 Rejected equipment. When equipment had been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer design, corrective action shall be performed on all equipment delivered under the contract.

4.10.1.3 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4.1 to ensure that it conforms to the requirements of this inspection. After reconditioning, reinspection and retesting, the equipment may be delivered as part of the contract quantity provided it conforms to all terms and conditions of the contract for acceptance.

4.10.1.4 Reliability test procedures. The reliability test shall be conducted using the procedures, schedules and performance limits specified herein, as a minimum.

4.10.1.4.1 Input line voltage. Input line voltage shall be as specified in Test Level C of MIL-STD-781. The input voltage shall be cycled in accordance with MIL-STD-781.

4.10.1.4.2 Test frequencies. During the reliability tests, at least one frequency shall be tested in each band as shown in TABLE IV:

TABLE IV. Test frequencies.

F1	2.0015 MHz
F2	5.3340 MHz
F3	10.6655 MHz
F4	29.9990 MHz

4.10.1.4.3 Test schedule. The test schedule of TABLE V, including the input line voltage schedule of 4.10.1.4.1, shall be conducted. The test shall consist of three 8-hour schedules per day, the first being manned during which performance measurements shall be made. The second two may be unmanned and pertinent data will be automatically recorded only to the extent that instrumentation permits.

TABLE V. Test schedule.

Manned Schedule	Unmanned Schedule
0-2 hours Transmitter ON 2-3 hours (See 4.10.1.4.4 Performance Measurements) 3-4 hours Transmitter OFF 4-6 hours Transmitter ON 6-7 hours Transmitter measurements 7-8 hours Transmitter OFF	0-3 hours Transmitter ON 3-4 hours Transmitter OFF 4-7 hours Transmitter ON 7-8 hours Transmitter OFF

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

4.10.1.4.3.1 Manned transmitting schedule. The mode of operation and frequency transmi for each of the transmitting ON periods of the manned schedules shall occur in the following sequence.

TABLE VI. Transmitting schedules.

Day 1/	Frequency	Mode of Transmission
1	F1 - First 4 hours of manned schedule F2 - Remainder of the day	CW (A1) 50 watts average power keyed at a rate of 32 baud, transmitting 10 minutes out of each 15 minute period.
2	F3 - First 4 hours of manned schedule F4 - Remainder of the day	AM (A3h) 25 watt carrier modulated 30-50% at 1000 Hz, transmitting for 10 minutes out of each 15 minute period.
3	F1 - First 4 hours of manned Schedule F2 - Remainder of the day	ISB (A3b) 100W PEP two-tone modulation per sideband, transmitting 10 minutes out of each 15 minute period.
4	F3 - First 4 hours of manned schedule F4 - Remainder of the day	RATT (A7j) 50 W average power keyed from space to mark at an 80 baud rate transmitting 10 minutes out of each 15 minute period.

1/ The same sequence shall be used on the next four days, except that the sequence of frequen shall be reversed, F4 and F3 on the fifth day, F2 and F1 on the sixth day, F4 and F3 on the seventh day, and F2 and F1 on the eighth day. The schedule shown in TABLE VI shall be repea as necessary.

4.10.1.4.3.2 Unmanned transmitting schedule. The mode of transmission during the unma transmitting periods shall be ISB (A3b) using voice tape modulation for both USB and LSB, 100 watts PEP total of both channels, transmitting for 10 minutes out of each 15 minute peri The test frequency to be used during each unmanned period shall be the last frequency utiliz during the previous manned period (see 4.10.1.4.3.1).

4.10.1.4.4 Minimum performance limits. During each of the specified periods of perfor ance measurements, the contractor shall perform the necessary tests to assure that equipment performance is not less than the following:

a. CW (A1) mode
   1. Keying rate shall be at least 32 baud
   2. Output shall be at least 50 watts +1.5 dB, -1.0 dB average power into a 50 ohm load in key down condition
   3. Keyed wave shall be free from transients and splatter
b. AM (A3h) mode
   1. No adjustment of the internal carrier reinsert control shall be made dur the test providing carrier power level remains within 25 watts +1.5 dB, -1.0 dB into a 50-ohm load. If carrier power variations exceeds above limits, sufficient adjustment range of the carrier reinsert control to adjust the carrier power to 25 watts must be demonstrated.
   2. It shall be possible to apply an audio tone of 1000 Hz to the upper side input and adjust the level so that the sideband output equals the carrie (25 watts +1.5 dB, -1.0 dB). Without changing any controls on the trans mitter, the sideband power shall not vary more than 3 dB when the sideba frequency is changed from 300 to 3500 Hz and the input level is maintain constant at the level required for equal sideband and carrier output at 1000 Hz input.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

3. Under the test conditions of step 2, the intermodulation distortion products shall be at least 27 dB below the carrier

c. SSB (A3a) mode
1. With a standard two-tone input applied to the upper sideband local input, the output of the RF amplifier shall be at least 100 watts PEP +1.5 dB, -1.0 dB into a standard dummy load.
2. With modulation frequencies of 1000 and 1625 Hz, the intermodulation distortion shall be not less than 27 dB below either tone of a two tone envelope at 100 watt PEP (100 watts PEP +1.5 dB, -1.0 dB), when measured into a standard dummy load.
3. Opposite sideband suppression shall be at least 60 dB when a 50 mV, 1000 Hz tone to the upper sideband local input is adjusted for 25 watts output into a standard dummy load.
4. Carrier suppression shall be adjustable to at least 50 dB below a sideband tone of 1000 Hz having an input level of 50 mV applied to the upper sideband local input.
5. Repeat 1 through 4 with the inputs applied to the lower sideband local input.

d. RATT (A7j) mode
1. Keying rate shall be at least 80 baud
2. Frequency shall be ±425 Hz ±5 percent shift from 2000 Hz with input neutral keying of 120 VDC 60 mA nominal

e. Frequency stability: The drift of the reference oscillator (internal frequency standard) shall be not worse than one part in 10^7 per day nor one part in 10^6 per month.

f. CU-937( )/UR tests: Power output, VSWR, and rough and fine tuning characteristics shall be verified in accordance with MIL-A-23547.

4.10.1.5 Burn-in. The contractor may elect to operate the reliability test samples for a certain time prior to the initiation of reliability demonstration testing, in addition to the burn-in period specified in 3.13.2. Any such burn-in applied to test samples must be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered nonrelevant but shall be recorded and reported. Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the reliability demonstration test samples.

4.10.2 Equipment burn-in. All deliverable equipments and spares shall be subjected to the temperature burn-in requirements of 3.13.2. The burn-in shall be performed at a stabilized 50°C ambient temperature with 1.0g ±10 percent vibration applied at least 10 minutes out of each operating hour for a cumulative operating time of at least 24 hours. Failures occurring during the last 24 hours shall be repaired and burn-in resumed for a minimum of 24 hours without further failure.

4.10.2.1 Equipment operation during burn-in. Equipment operation need not be continuous as long as the accumulated operating time conforms to conditions of 4.10.2. The equipment shall be operated in accordance with 3.13.2.

4.10.2.2 Performance monitoring. Equipment shall be checked for proper performance at least twice every 24 hours and just prior to end of burn-in.

4.10.3 Failure analysis. All failures shall be analyzed and the results of analysis shall be retained by the contractor and be available for Government review as required.

4.11 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:48Z
Check the source to verify that this is the current version before use.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order).

5.1 Preservation, packing, packaging and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The Radio Transmitter AN/URT-24( ) is intended for general purpose voice, CW, and radio teletypewriter communications, and may be used for surface ship, submarine, and fixed shore installations.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title number and date of this specification
b. Quantity of first article sample units required if other than specified in 4.3
c. Government furnished material (see 3.2)
d. Interchangeability requirement, if required (see 3.3)
e. Quantity of accessory items required (see 3.4.1)
f. Rack mounting brackets, as required (see 3.15.1.2.1)
g. Number of extender cables required (see 3.15.1.35)
h. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7.104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DD form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7.104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirements	Applicable DID
4.5	Test Plans	DI-T-3709
4.5	Test Procedures	UDI-T-22710

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of at least one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Nomenclature. Nomenclature referred to herein and those assigned to the GFE units may be subject to change on any contract resulting in equipment produced in accordance with this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.

6.5 Environmental inspection. Approval to ship may be with held at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.6 Standard definitions.

6.6.1 Local input. The term local input in this specification indicates that control and modulating functions for the transmitter are being provided by the handset (or other appropriate device connected to the handset receptacle on the front panel) or by the hand key on the front panel.

6.6.2 Remote inputs. The term remote inputs in this specification indicates that control and modulating functions will be provided at the rear of the transmitter case.

6.6.3 Standard two-tone audio input. Standard two-tone audio input is defined as two equal amplitude audio signals, 1000 Hz and 1625 Hz, with a level of 150 millivolts per tone at the remote input terminals.

6.6.4 Standard amplifier driving source impedance. This impedance shall be 50 ohms resistive.

6.6.5 Standard dummy load. This impedance shall be a 50 ohm resistance of sufficient power capacity to dissipate at least 100 watts of average RF power without change of its electrical characteristics.

6.6.6 Hum signals. Hum signals are any output products resulting from modulation at the primary power line frequency or power supply conversion frequencies or harmonics of these frequencies.

6.6.7 Standard single-tone audio input. Standard single-tone audio input is defined as a 1000 Hz tone, with a level of 150 millivolts at the remote input terminals.

6.6.8 Failure. Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing within the specified operational requirements.

6.7 Changes for previous issue. The extent of changes (deletions, additions, and so forth) preclude the annotation of the individual changes from the previous issue of this document.

Preparing activity:
NAVY-EC
(Project 5820-N695)

Source: https://assist.dla.mil -- Downloaded: 2026-01-19 18:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-10T18:48Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-19T18:48Z
Check the source to verify that this is the current version before use.